

Q1. Intelligent Customer Feedback Analysis

CODE

```
def preprocess(text):
    text=text.lower().strip()
    text=text.replace(".", "")
    text=text.replace(",", "")
    text=text.replace("!", "")
    text=text.replace("?", "")
    return text
def analyze(text):
    positive=["good", "excellent", "great", "happy", "fast", "love"]
    negative=["bad", "poor", "slow", "late", "worst", "hate"]
    p = 0
    n = 0
    words =text.split()
    for word in words:
        if word in positive:
            p += 1
        elif word in negative:
            n += 1
    print("Positive Keywords:", p)
    print("Negative Keywords:", n)
    if p > n:
        print("Category: Positive")
    elif n > p:
        print("Category: Negative")
    else:
        print("Category: Neutral")
num = int(input())
for i in range(num):
    feedback = input()
    feedback = preprocess(feedback)
    analyze(feedback)
```

INPUT

```
2
The product is Good and Fast.
Worst service, very slow!
```

OUTPUT

```
Positive Keywords: 2
Negative Keywords: 0
Category: Positive
Positive Keywords: 0
Negative Keywords: 2
Category: Negative
```

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Q2. Comprehensive Password Strength Checker

CODE

```
def password_check(p):  
    u=0  
    l=0  
    d=0  
    s=0  
    for i in p:  
        if i.isupper():  
            u += 1  
        elif i.islower():  
            l += 1  
        elif i.isdigit():  
            d += 1  
        else:  
            s += 1  
    if len(p) == 0:  
        print("Password is Empty")  
    elif len(p) < 6:  
        print("Strength: Weak")  
    elif len(p) ≥ 8 and u > 0 and l > 0 and d > 0 and s > 0:  
        print("Strength: Very Strong")  
    elif u > 0 and l > 0 and d > 0:  
        print("Strength: Strong")  
    else:  
        print("Strength: Medium")  
pwd = input()  
password_check(pwd)
```

INPUT

OUTPUT

Q3. Structured Server Log Error Analyser

CODE

```
def check_log(logs):
    e=0
    w=0
    ans=[]
    for i in logs:
        if i.startswith("ERROR") or i.find("ERROR") != -1:
            e += 1
            if i not in ans:
                ans.append(i)
        elif i.startswith("WARNING") or i.find("WARNING") != -1:
            w += 1
    print("Error Count:", e)
    print("Warning Count:", w)
    print("Unique Errors:")
    for i in ans:
        print(i)
n = int(input())
logs = []
for i in range(n):
    logs.append(input())
check_log(logs)
```

INPUT

```
5
ERROR Disk Full
INFO Server Started
WARNING Low Memory
ERROR Disk Full
ERROR File Missing
```

OUTPUT

```
Error Count: 3
Warning Count: 1
Unique Errors:
ERROR Disk Full
ERROR File Missing
```

Q4. Email Domain Frequency Analyser

CODE

```
def domain_count(mail):
    d = {}
    for i in mail:
        if "@" in i:
            x = i.split("@")
            if len(x) == 2:
                dm = x[1]
                if dm in d:
                    d[dm] += 1
                else:
                    d[dm] = 1

    m = 0
    name = ""
    for i in d:
        print(i, ":", d[i])
        if d[i] > m:
            m = d[i]
            name = i
    print("Most Common Domain:", name)

n = int(input())
mail = []
for i in range(n):
    mail.append(input())
domain_count(mail)
```

INPUT

```
5
abc@gmail.com
xyz@yahoo.com
pqr@gmail.com
test@gmail.com
hello@yahoo.com
```

OUTPUT

```
gmail.com : 3
yahoo.com : 2
Most Common Domain: gmail.com
```